

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 1

CHAPTER 1: METALLURGY

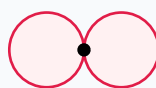
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 1
Chapter Name:	Metallurgy

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)



p orbital (px)

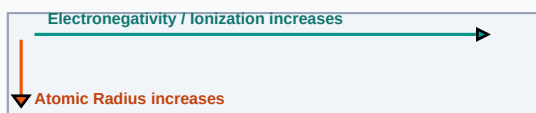
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Ore	A mineral from which a metal can be extracted commercially and profitably.
Gangue	The non-metallic and stony impurities present along with ores in mines (e.g. silica, clay).
Calcination	The process of heating an ore in the absence of air or in limited supply, decomposing carbonates and hydroxides.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

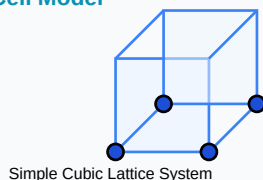
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Froh flotation activation: Xanthates select sulfide particles
- Zone refining: impurity solubility liquid > solid phase

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

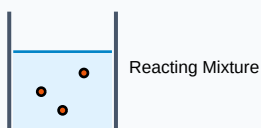
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Differentiate between roasting and calcination.

Step-by-step Solution:

Roasting heats sulfide ores in excess oxygen to form oxides (e.g. $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$). Calcination heats carbonate or hydrated ores in absence of air to expel volatile water/ CO_2 (e.g. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$).

Titration Beaker & Solute Precipitation



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

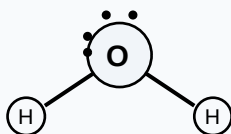
HOTS Challenge 1: Explain the thermodynamic principles of metallurgy using Ellingham diagram curves.

Analytical Solution Strategy:

Plots Gibbs energy change versus temperature for oxide formation.

A metal can reduce the oxide of another metal if its curve lies below the other metal's curve at that temperature, as the net free energy change $\Delta_r G$ of the combined reaction is negative.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

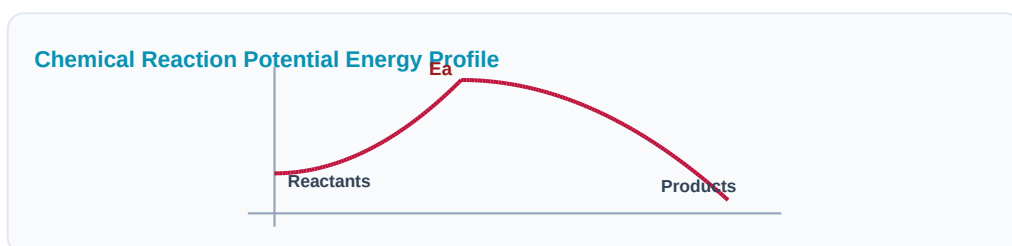
Q1 (1-Mark): Describe the froth floatation method for concentrating sulfide ores.

Q2 (2-Mark): Explain the extraction of gold using cyanide leaching.

Q3 (2-Mark): Describe Mond process for refining Nickel with equations.

Q4 (5-Mark): List the ores of aluminium, iron, and copper.

Q5 (5-Mark): What is slag? How is it formed in blast furnaces?



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Aviation Alloys

Hall-Heroult smelting isolates pure aluminium, structural metal for plane frames.

Application Case: Electronics

Zone refining purifies silicon crystals to 99.999999% purity for CPU microchips.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

